

Pavel Marius Florian Adrian

Grupa 22

Laborator 9

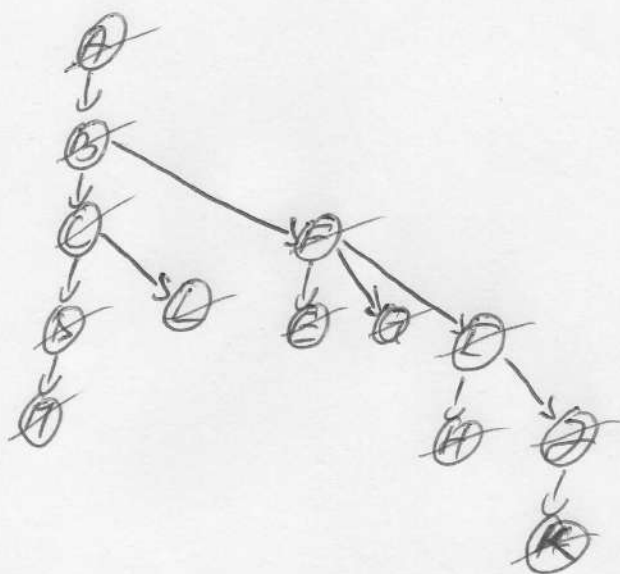
1.

Nod	A	B	C	D	E	F	G	H	I	J	K	L	M
Grad	0	1	2	3	2	2	2	1	1	2	3	2	3
Interm		0	1	2	1	1	1	0	0	1	2	1	2
			0	1	0	0	0			0	1	0	1
				0							0		0

Q: A, B, F, E, I, H, J, C, G, K, L, M

L: A, B, F, E, I, H, J, C, G, K, D, L, M

A	B	C	D	E	F	G	H	I	J	K	L	M
0	0	0	0	0	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1	1	1	1



Stivă:

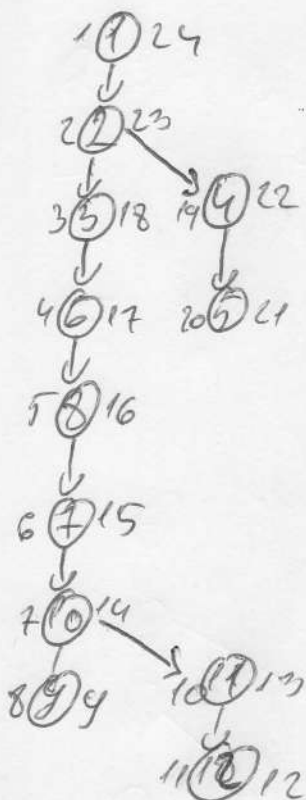
A
B
F
E
G
H
K
L
M

Se sterge nodurile de pe stivă:
A, B, F, E, J, K, H, G, E, C, L, D, M

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2. Algoritmii lui Kosaraju:

1 2 3 4 5 6 7 8 9 10 11 12
~~0~~ ~~0~~ ~~0~~ ~~0~~ ~~0~~ ~~0~~ ~~0~~ ~~0~~ ~~0~~ ~~0~~ ~~0~~ ~~0~~
 + + + + + + + + + + + + +



1-vizitat 0 d[2]

2-vizitat 2 d[2, 5, 4]

5-vizitat 5

4-vizitat 4

8-vizitat 8 d[7, 8, 9, 10, 11, 12]

7-vizitat 7

9-vizitat 9

10-vizitat 10

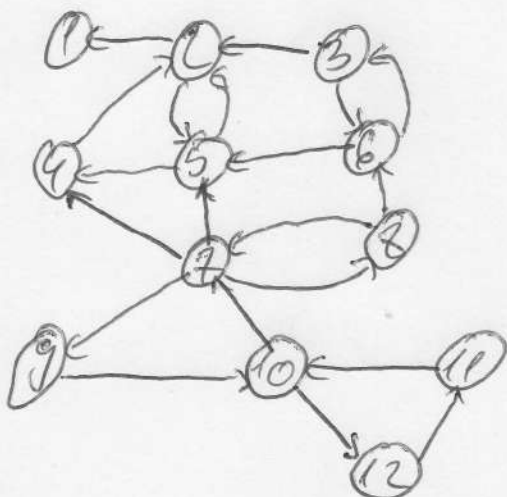
12-vizitat 12

11-vizitat 11

3-vizitat 3 d[3, 6]

6-vizitat 6

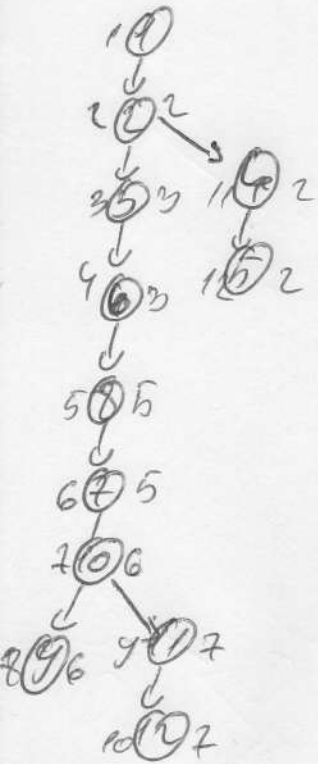
Există 4 componente tare conexe



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2. Algoritmul lui Tarjan

| | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|----|----|----|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |



Stivă

| |
|----|
| 12 |
| 11 |
| 9 |
| 10 |
| 7 |
| 8 |
| 6 |
| 3 |
| 2 |
| 1 |

| |
|---|
| 5 |
| 4 |
| 2 |
| 1 |

$$low_8 = \min(premum_3, premum_7) \\ = \min(8, 6) = 6$$

$$preum_{11} = 8 + 1 = 9$$

$$preum_{12} = 5 + 1 = 10$$

$$low_{12} = (\text{preum}_{12}, \text{preum}_{10}) \\ = \min(10, 7) = 7$$

$$low_{11} = (\text{preum}_{11}, low_{12}) = 7$$

$$low_{10} = \min(\text{preum}_{10}, low_9, low_{11}) = 6$$

$$low_7 = \min(\text{preum}_7, low_{10}, \text{preum}_8) = 5$$

$$low_8 = 5 \quad low_4 = 2$$

$$low_6 = 3 \quad low_2 = 2$$

$$low_5 = 2 \quad low_1 = 1$$

$$low_3 = 3$$

Există 4 componente tare conexe

{12, 11, 9, 10, 7, 8}

{6, 3}

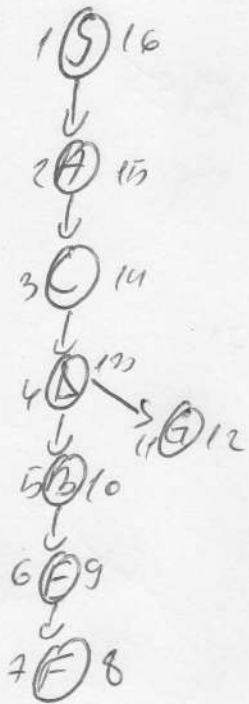
{5, 4, 2}

{1}

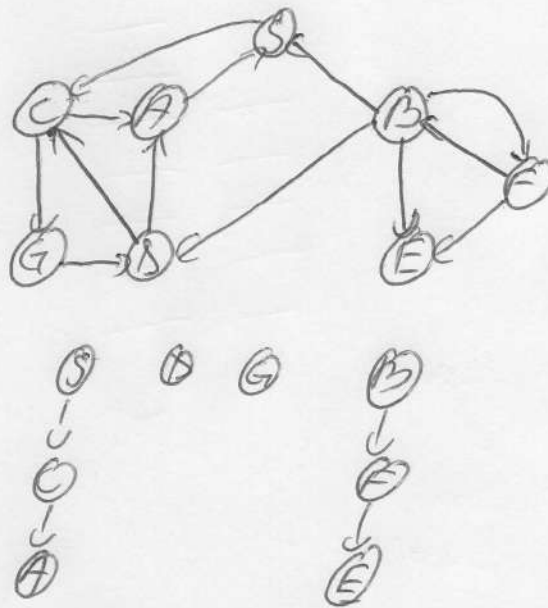
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3. Algoritmul lui Kosaraju

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| S | A | B | C | D | E | F | G |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |



Grat transpus:



Există 4 componente tare conexe:

$\{S, C, A\}$

$\{D\}$

$\{G\}$

$\{B, E, F\}$

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3. Algoritmul lui Tarjan



Stivă:

| |
|---|
| G |
| F |
| E |
| B |
| D |
| C |
| A |
| S |

$$\text{low}_F = (\text{premium}_F, \text{premium}_B) \\ = \min(7, 15) = 5$$

$$\text{low}_E = \min(\text{premium}_B, \text{low}_F) \\ = \min(6, 5) = 5$$

$$\text{low}_B = \min(\text{premium}_B, \text{low}_E) \\ = \min(5, 15) = 5$$

$$\text{low}_G = \min(\text{premium}_G, \text{premium}_D) \\ = \min(8, 4) = 4$$

$$\text{low}_D = 4 \quad \text{low}_A = 1$$

$$\text{low}_C = 1 \quad \text{low}_S = 1$$

Există 4 componente tare conexe:

$\{B, E, F\}$

$\{G\}$

$\{D\}$

$\{C, A, S\}$